AYMANE AIT DADS

Software Engineer Intern - Kernels | C/C++ · PyTorch · Hardware Optimization

Aymane.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

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Hiring Team - Software Engineer Intern - Kernels
QUADRIC PTY LTD · Burlingame, CA, USA

Dear Quadric Hiring Team,

Your posting calls for someone who can **develop AI/LLM kernels** and **profile kernel performance across compute, data, and parallelism** - exactly the intersection I have been building toward. Fine-tuning **CLIP ViT-L/14 (428M parameters)** on 250K samples required me to reason about GPU memory constraints, operator-level efficiency, and inference throughput under real hardware limits, not just model accuracy - ranking #1 on the EURECOM leaderboard as a result.

To maximize **hardware utilization** within a fixed T4 GPU budget, I applied **FP16 mixed precision with GradScaler** and designed a **10-view test-time augmentation** pipeline, directly trading compute headroom for inference robustness - the same bottleneck-driven thinking your kernel engineers apply to the Quadric GPNPU stack. Separately, building a **transformer autoencoder** for unsupervised anomaly detection required profiling feature representations - Mel vs. MFCC - to identify where compute was wasted, achieving **AUC 0.80** with no labeled data.

Quadric's bet that a single GPNPU can execute both **NN graph code and C++ DSP control code** without offloading to a host CPU is an architectural position most NPU vendors have not been willing to take - and it is precisely the layer where kernel-level optimization becomes the differentiating factor. I would welcome the chance to contribute to that stack from your Burlingame office this summer.

Sincerely,

AYMANE AIT DADS